United International University (UIU)
Dept. of Computer Science & Engineering (CSE)
Final Exam, Trimester: Fall 2025

Course Code: CSE 113/EEE 2113; **Course Title:** Electrical Circuits

Total Marks: 40; **Duration:** 2 hour

1. **Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.**
2. **You must write your serial Number on the top of your Exam's Answer Sheet.**

Question 1: Answer all the questions. (15 Marks)

Rashed and Tanvir built a linear DC circuit with a variable load resistor connected across terminals a–b. Rashed claims that connecting a 4 Ω resistor across terminals a–b will ensure maximum power transfer. Tanvir disagrees, stating that maximum power transfer occurs when a 3.5 Ω resistor is connected across terminals a–b. Now, answer the following questions: [8+2+2+2+1] **CO3**

i) **Determine** the Thevenin equivalent circuit seen from terminals a–b.
ii) **Calculate** the current flowing through the load resistance when the load is 4 Ω, and then when the load is 3.5 Ω.
iii) **Calculate** the power delivered to the load when following Rashed's advice, and then calculate the power delivered to the load when following Tanvir's advice.
iv) **Explain** when the maximum power is delivered to the load and whose claim (Rashed's or Tanvir's) is correct.
v) If you want to connect a bulb between a-b for maximum power, **what** could be the rating of this bulb? Rating of a bulb means the typical operating voltage of a bulb and its corresponding power.

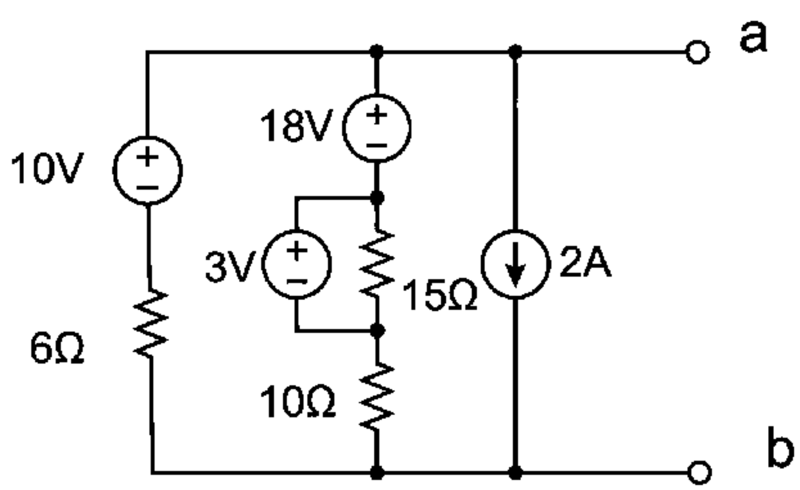

Figure 1

<u>Question 2: Answer all the questions. (15 Marks)</u>

For the circuit shown in **Figure 2**, answer the following questions: [5+4+4+2] **CO4**

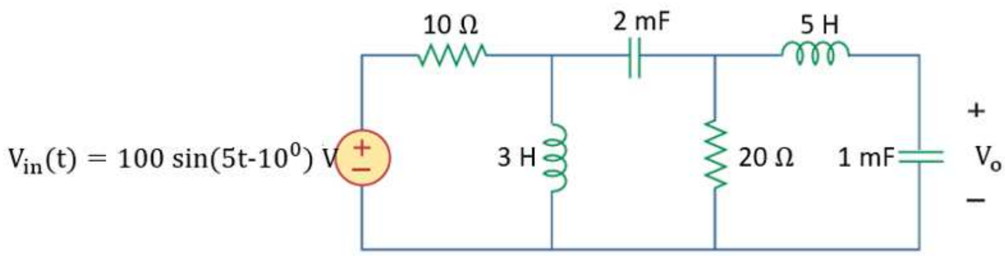

Figure 2

i) **Redraw** the circuit in the phasor domain and **determine** the equivalent impedance as seen by the sinusoidal voltage source.

ii) **Determine** $V_o(t)$ using CDR or VDR.

iii) **Determine** the phase angle difference between the input, $V_{in}(t)$, and output voltage, $V_o(t)$, and find by how much the output voltage is lagging.

iv) **Calculate** the average real power absorbed by the 20ohm resistor.

<u>Question 3: Answer all the questions. (10 Marks)</u>

A 10 Ω resistor receives a **periodic voltage v(t)** across it. The waveform is shown below. [4+4+2] **CO4**

i) **Derive** the mathematical expression of v(t).

ii) **Determine** the rms value of v(t).

iii) **Determine** the power absorbed by the resistor.

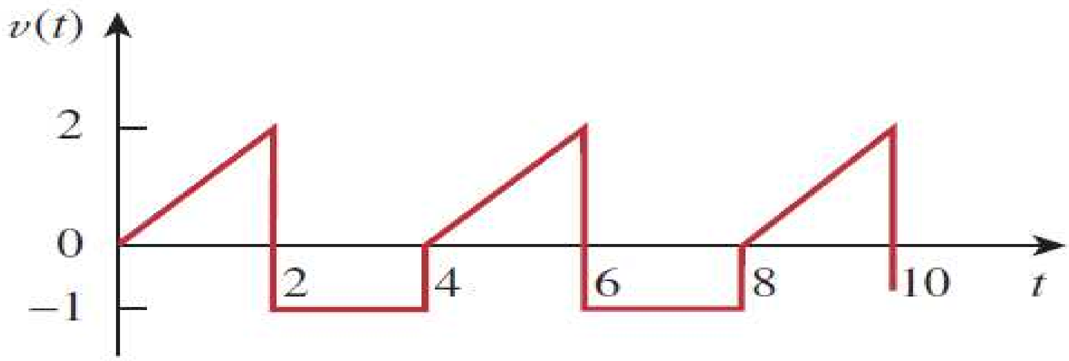

Figure 3